SECTION 2-1

Big Data and Data Engineering

The University of Tokyo
Center for Research and Development on Transition to Modern Education

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Overview

Understanding the progress of ICT and Big Data Systems:

01 Socio-Technical Infrastructure Background

Understand the technological background and infrastructural advancements (e.g., broadband and cellular evolution) that made data engineering possible.

02 Multi-Domain Industrial Applications

Explore practical data-driven case studies across retail, manufacturing, logistics, and primary industries (agriculture & fishery).

Table of Contents

I.	Conceptual Definition and Data Proliferation	pp. 4–5
II.	Four Key Challenges of Big Data Leverage	p. 6
III.	Enabling Background: ICT & Hardware Infrastructure	pp. 7–13
IV.	Multi-Domain Case Studies (Retail, Logistics, Industrial)	pp. 17–26
V.	Socio-Technical Data Classifications & Primary Sector	pp. 27–31

Defining Big Data

Massive Digital Telemetry

At its core, Big Data refers to massive volumes of digital information that are generated, formatted, and propagated seamlessly across the internet.

The Proliferation of Behavioral Logs

Typical examples include social network interactions (SNS), search histories, and purchase telemetry on e-commerce platforms or physical POS retail networks.

Natural & Experimental Observations

Big data extends to scientific and natural realms, such as high-frequency meteorological data, seismic wave sensors, and particle physics experiment outputs.

> ***"Big data: The next frontier for innovation, competition, and productivity."***
>
> — McKinsey Global Institute

Source: [McKinsey Report (2011)](https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/big-data-the-next-frontier-for-innovation)

The Proliferation of Data Volume

Ubiquitous Data Generation

Modern society is constantly surrounded by massive streams of continuous data. Every micro-action contributes directly to the expansion of the digital universe.

High-Frequency Micro-Actions (Per Minute)

Statistical studies reveal staggering levels of data creation every single minute: millions of media views, hundreds of millions of emails, and hundreds of thousands of uploads.

Accelerating Big Data Creation

This organic, user-driven behavior across smartphones, applications, and social platforms acts as the primary catalyst accelerating big data generation globally.

YouTube Video Views	4.5 Million
Emails Transmitted	188 Million
Instagram Stories/Posts	335,000

Every minute metrics based on 2019 census data. [Domo Data Blog](https://web-assets.domo.com/blog/wp-content/uploads/2019/07/data-never-sleeps-7-896kb.jpg)

Four Key Challenges of Big Data Leverage

01

Massive Storage Capacity

How to reliably and cost-effectively ingest, store, and partition petabyte-scale unstructured datasets over time.

02

High-Performance Processing

Achieving low latency and high throughput when filtering, sorting, or joining massive datasets in distributed architectures.

03

High-Value Knowledge Extraction

Leveraging statistical models, machine learning, and semantic pipelines to convert raw telemetry into valuable business insights.

04

Security & Privacy Protection

Establishing robust cryptographic frameworks, anonymization, and access control models to safeguard sensitive user data.

Enabling Background: ICT Progress

Rapid Progress of ICT

The rising prominence of Big Data and modern Data Engineering is unequivocally driven by the massive advancements in Information and Communications Technology (ICT).

Explosive Data Ingestion Capabilities

High-capacity telecommunication infrastructure has drastically boosted data collection pipelines. Simultaneously, user behaviors (e.g. SNS) have moved raw data production completely online.

Focus on Network Proliferation

To understand this evolution, we must analyze the rapid penetration of high-speed broadband and high-bandwidth wireless cellular networks globally.

Data Collection Lifecycle

1. Telemetry Generation → 2. High-Speed Transit → 3. Central Aggregation

Continuous ingestion powered by wireless & wired fiber networks

Figure: Structural flow of modern network data ingestion.

The Evolution of Broadband Contracts

Wired & Wireless Proliferation

The structural metrics show a rapid transition toward high-speed, high-performance bandwidth lines on both fixed and mobile networks.

Fixed Broadband Growth

Fixed-line infrastructure migrated rapidly from copper DSL models to ultra-fast fiber FTTH (Fiber-to-the-Home) networks, establishing Gbps-scale household access.

Explosive Mobile Ingestion

The adoption of broadband mobile contracts (such as LTE and BWA) surpassed fixed networks by orders of magnitude, cementing continuous, high-volume mobile telemetry.

Infrastructure Glossary	
FTTH	Fiber-to-the-Home
DSL	Digital Subscriber Line
CATV	Cable Television Broadband
BWA	Broadband Wireless Access
LTE	Long Term Evolution (4G)

Source: [Ministry of Internal Affairs and Communications (MIC) White Paper (2020)] (https://www.soumu.go.jp/johotsusintokei/whitepaper/ja/r02/html/nd252210.html)

Mobile Network Traffic Explosion

Socio-Technical Traffic Growth

Paralleling the broadband contract proliferation is the exponential surge in actual mobile network traffic transiting through regional telecommunication cellular towers.

High-Frequency Mobile Adoption

The rising integration of high-definition streaming, cloud telemetry, app distribution, and continuous GPS location tracking has generated astronomical traffic growth.

The Data Engineering Mandate

Handling these soaring multi-terabit network loads made the development of highly scalable distributed computing frameworks and database engines an absolute necessity.

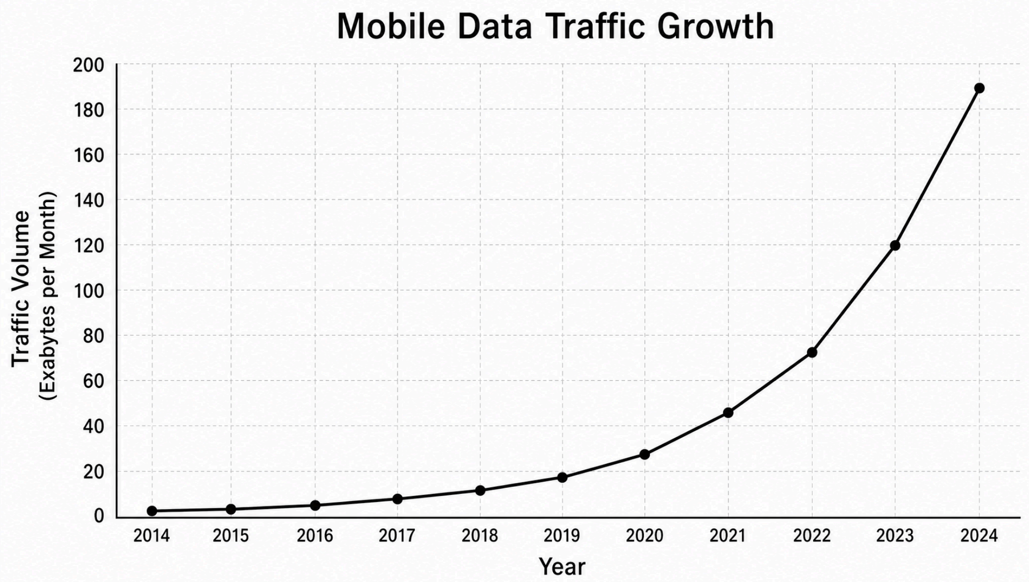

Figure: Rapid exponential growth curve of mobile communication traffic. [MIC Whitepaper] (https://www.soumu.go.jp/johotsusintokei/field/data/gt010602.pdf)

Storage Hardware Expansion Trends

Solving the Ingestion Bottleneck

The first primary challenge of big data leverage—retaining massive volumes of information—was systematically resolved by hardware innovations in storage density.

Magnetic & Solid State Evolution

Over recent decades, HDD (Hard Disk Drive) density and DRAM (Dynamic Random Access Memory) chip capacity have followed an aggressive exponential growth curve.

Foundations of Distributed Databases

This rapid, hardware-driven drop in cost-per-gigabyte laid the indispensable foundation for building modern, distributed database architectures.

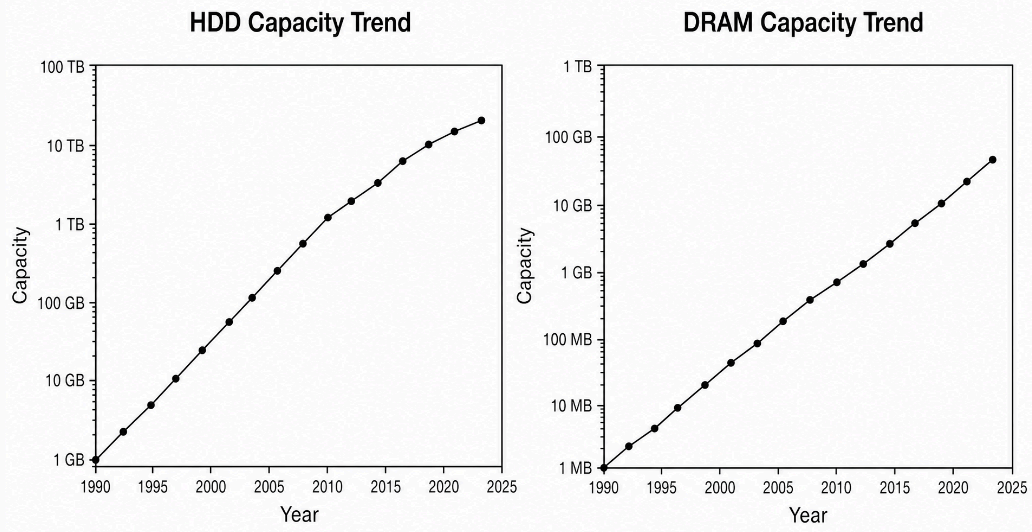

Figure: Historical capacity expansion trends of HDD and DRAM hardware. [NHK Science & Technology Research Labs](https://www.nhk.or.jp/strl/publica/rd/181/2.html)

Big Data Case Studies: POS Analytics

Physical Point-of-Sale Telemetry

Recording transactions directly inside physical retail outlets (POS) captures rich behavioral metadata, acting as a crucial component of modern marketing analytics.

Market Basket & Supply Synergy

Analyzing co-occurrence probabilities of purchase items enables optimal shelf layouts. Modeling temporal variations allows adaptive supply chain planning based on weather or localized events.

Direct Campaign Measurement

Analyzing exact purchase reaction logs after marketing campaigns allows food & beverage enterprises to drastically trim wasteful advertising costs while expanding gross margins.

Market Basket Analysis

Mining retail purchase transactions to find association rules (e.g., "People buying Item X are 82% likely to buy Item Y").

Case Study: One-to-One Marketing Campaign Efficiency. [Liskul Retail Analytics](https://liskul.com/one-to-onemarketing-13202)

Big Data Case Studies: EC Marketing

Granular Behavioral Logs

Unlike physical stores, digital actions on e-commerce (EC) platforms generate continuous, granular clickstream logs and behavioral records per individual user.

Cookie-Based Identity Tracking

Utilizing client-side cookie identifiers and past purchase histories allows operators to run personalized, real-time campaign optimizations rather than mass marketing.

One-to-One Marketing Paradigm

This systematic strategy focuses on targeting users individually with tailored recommendations or related-product promotions, which boosts conversion rates.

One-to-One Personalization

Tracking cookies and user clickstream dynamics to dynamically serve relevant content and product recommendations in real time.

Concepts: E-commerce behavioral telemetry and personalized targeting.

Two Paradigms of Recommendation

Value of Recommendation Systems

Recommendation systems represent one of the most commercially successful implementations of big data, serving as a primary driver of digital platform engagement.

Collaborative Filtering

Models user preferences based entirely on aggregate purchase histories of other similar users. It requires no demographic metadata but suffers from cold-start problems with new items.

Content/Attribute-Based Filtering

Utilizes metadata profiles of users and product attributes directly. While highly effective at recommending brand-new releases, it cannot function if attribute profiles are missing.

Collaborative Filtering

Aggregates purchase history of similar users to forecast item ratings. Eliminates metadata dependency.

Attribute-Based Filtering

Maps explicit customer properties (age, location) against product descriptors. Ideal for cold-start items.

Figure: Comparing collaborative filtering vs. attribute-based approaches.

Amazon: Item-to-Item Collaborative Filtering

Personalization as the Core Strategy

Amazon.com leverages its recommendation engine not merely as an add-on, but as a central framework to completely personalize the user's web storefront experience.

Item-to-Item Collaborative Filtering

First implemented in 1998, this classic algorithm pairs purchased items with closely-related inventory. This approach scales seamlessly to millions of concurrent customers.

Hybrid Offline/Online Computation

Computes complex product similarity tables entirely offline. The online component performs simple key-value lookups based on a user's active cart, ensuring sub-second response times.

Hybrid Architecture

Offline	➔ Heavy Item Similarity Matrix
Online	➔ Active Cart User Query
Execution	➔ Rapid Table Lookup & Rank
Total CTR Drive	**~30% of Site Views**

Source: [Linden, Smith, & York (2003)](https://ieeexplore.ieee.org/document/1167344) & [Smith & Linden (2017)](https://ieeexplore.ieee.org/document/7924241)

YouTube: Co-viewing Recommendation Model

Defining Video Relatedness

YouTube maps video associations based on co-viewing patterns—identifying which subsequent videos are watched after viewing an initial Video A.

Precomputed Association Sets

The algorithm pre-calculates the Top-N related videos for each entry, forming a precomputed related video set R(A) to ensure instantaneous serving latency.

Recursive Recommendation Graph

The live system cross-references active user histories with these sets, expanding recommendations to include secondary degrees of association, such as related videos of R(A).

Recursive Mapping Flow

- **Active Seed** ➔ User watches Video A
- **Primary Set R(A)** ➔ Precomputed co-viewed associations
- **Secondary Expansion** ➔ Related nodes within R(R(A))

Source: [Davidson et al., ACM Recommender Systems (2010)] (https://dl.acm.org/doi/10.1145/1864708.1864770)

Netflix: Multi-Algorithm Recommendation Matrices

High CTR Conversion Drive

Approximately 80% of total viewing hours on Netflix are driven directly by personalized recommendations, underscoring the critical business value of algorithmic personalization.

The Unified Grid Interface

Rather than using a single algorithm, Netflix serves a matrixed interface where each horizontal row represents an entirely different specialized recommender model.

Multi-Dimensional Scoring

Separate algorithms run concurrently in real time, sorting specific categories, trending releases, and genre preferences adjusted to user demographic profiles.

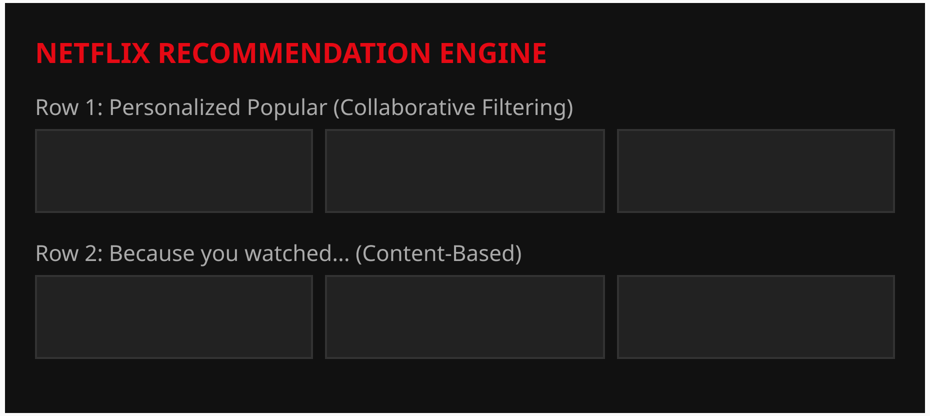

Figure: Mock of row-based algorithm mapping in the Netflix interface. [Gomez-Uribe & Hunt (2016)] (https://dl.acm.org/doi/10.1145/2843948)

Big Data Case Studies: Human Mobility

Mapping Spatio-Temporal Flows

Measuring, predicting, and optimizing human movement dynamics in physical space represents a highly impactful area of big data application.

Optimizing Public Spaces

Commercial and public organizations capture visitor movement patterns within museums or retail layouts to dynamically restructure pathing, traffic signs, and exhibits.

Emergency Disaster Navigation

Connecting real-time crowding telemetry with digital signage systems allows authorities to dynamically route evacuation paths during catastrophic natural events.

Dynamic Crowd Routing

Synergizing continuous GPS/Wi-Fi signal mapping with municipal display panels to coordinate safe public evacuation pathing during emergencies.

Source: [AIST Artificial Intelligence Research Center](https://www.airc.aist.go.jp/achievements/ja/p-028.html) & [Nikkei XTech](https://xtech.nikkei.com/it/atcl/column/14/122600137/122600001/)

Big Data Case Studies: Machine Telemetry

Industrial IoT Telemetry

Just as human behavioral logs are parsed as big data, high-frequency physical telemetry and machine operational logs represent an identical paradigm shift.

Global Fleet Ingestion: Komatsu KOMTRAX

As an industry pioneer, Komatsu equips standard machinery fleets with KOMTRAX sensors, centralizing real-time positioning and mechanical strain logs over wireless networks.

Predictive Failure Analysis

Turbine power plant operators (e.g., IHI) aggregate massive operational telemetry, modeling baseline dynamics to predict critical anomalies and prevent physical failure.

Predictive Lifecycle Management

Centralizing global industrial assets, using continuous multivariate sensor streams to build prescriptive maintenance algorithms.

References: [Komatsu KOMTRAX](https://sanki.komatsu/komtrax/) & [IHI Lifecycle Operation Support](https://www.ihi.co.jp/powersystems/lifecycle/operation_support/index.html)

Big Data Case Studies: Syslog Diagnostics

System-Level Event Logs

High-volume information system logs represent critical, unstructured big data, parsed extensively to model complex infrastructure health.

Failure of Simple Rule-Based Alerts

Standard networks rely on hardcoded thresholds for hardware failures. However, they struggle to capture latent, pre-failure anomalies, traditionally relying on veteran experience.

Mapping Co-occurrence (Syslog Analysis)

A single anomaly triggers cascading logs across multiple network elements. Modeling these co-occurrence events enables highly precise diagnostic mapping.

Syslog Co-occurrence Matrix

Parsing massive system-level console message streams to detect temporal patterns preceding complex network faults.

Source: [T. Kimura, Syslog Analysis for Large-scale Network Failures (2015)](https://search.ieice.org/bin/summary.php?id=j98-b_9_823)

Socio-Technical Data Classifications

01

Image Telemetry

Advanced image processing algorithms for automated piloting systems, vehicle identification, smart food sorting, and deep-learning cancer detection.

02

Voice & Audio Streams

Continuous voice recognition (Apple Siri, Alexa) paired with advanced text-to-speech synthesis models used in automated news anchors.

03

Document & Natural Language

Leveraging neural machine translation models (Google GNMT) and deep generative neural networks to automate comprehensive article writing.

04

Sensor & Temporal Logs

High-frequency industrial IoT, syslog co-occurrence networks, GPS mobility tracks, and point-of-sale customer purchase history logs.

Industrial Innovations: Smart Agriculture

Spatio-Temporal Crop Monitoring

Analyzing farm conditions via multispectral satellite imagery and ground-level wireless sensor arrays enables highly efficient, localized crop management (e.g., viticulture).

Systematic Agricultural Collaboration

To scale productivity, national initiatives coordinate cross-industry databases, establishing unified data pipelines to sharing climate, soil, and yield metrics.

The WAGRI Integration Initiative

The establishment of the Agricultural Data Collaboration Platform (WAGRI) provides a highly unified API network that integrates public weather, geological, and agricultural telemetry.

WAGRI Platform

A national-scale collaborative API ecosystem integrating meteorological data, satellite crop maps, soil properties, and agricultural machinery logs.

Source: [Smart Agriculture Japan](https://smartagri-jp.com/smartagri/20) & [WAGRI Consortium Portal](https://wagri.net/ja-jp/)

Industrial Innovations: Smart Fisheries

Predictive Harvest Modeling

Building predictive harvest models based on historical catch records, meteorological conditions, and real-time ocean current sensors optimizes set-net fishery yields.

Deep Sea Resource Forecasting

Pelagic fleets (e.g., skipjack and tuna fisheries) aggregate massive oceanographical datasets, using sea surface temperatures to model fish school movement and map maritime resources.

Smart Aquaculture & Efficiency

Integrating water quality telemetry with automated computer vision feeders optimizes fish growth and operational workflows in modern aquaculture facilities.

Oceanographical Modeling

Combining sea temperatures, maritime weather logs, and sonar fish school coordinates to map global migratory pathways of pelagic fish species.

Source: [MIC Smart Fishery Initiative (2017)](https://www.soumu.go.jp/main_sosiki/joho_tsusin/top/local_support/ict/jirei/2017_013.html) & [Nissui Aquaculture Life-cycle](https://nissui.disclosure.site/ja/themes/144)

Summary: Big Data & Data Engineering

Socio-Technical Co-evolution

The exponential growth of big data is deeply intertwined with high-speed telecommunications, drop in hardware cost-per-gigabyte, and the proliferation of IoT networks.

High-Value Knowledge Extraction

Practical business success across e-commerce (Amazon), streaming (Netflix, YouTube), and logistics hinges on shifting from mass-marketing to precise, one-to-one algorithmic models.

The Broad Industrial Landscape

From high-frequency industrial IoT, Syslog anomaly mining, and GPS human mobility networks, to WAGRI agricultural APIs and predictive fisheries, data engineering actively optimizes our primary, secondary, and tertiary sectors.

The Engineering Paradigm

Transitioning from passive data storage to automated, real-time context-aware decision systems that actively optimize human societal capabilities.